



Faculty of Engineering and Technology Department

ARTIFICIAL INTELLIGENCE

ENCS3340 – Second Semester 25/26

Machine Learning Classification Project

Project 2

Prepared by : Heba Dababat

ID : 1230151

Section : 1

Dr : yazan

Date : 25-06-2026

Introduction

This report demonstrates the use of machine learning classification algorithms in Python using the Scikit-learn library. The project is organized into three main sections: In Part A, a Decision Tree classifier is developed and evaluated using the PlayTennis dataset. Part B extends the study by implementing a Naive Bayes classifier and comparing its performance with the Decision Tree. Finally, Part C applies a Decision Tree to a real-world dataset from the UCI Machine Learning Repository, specifically the Titanic dataset. All experiments were completed in Python 3 using Scikit-learn, pandas, matplotlib, and seaborn.

Part A (PlayTennis Decision Tree)

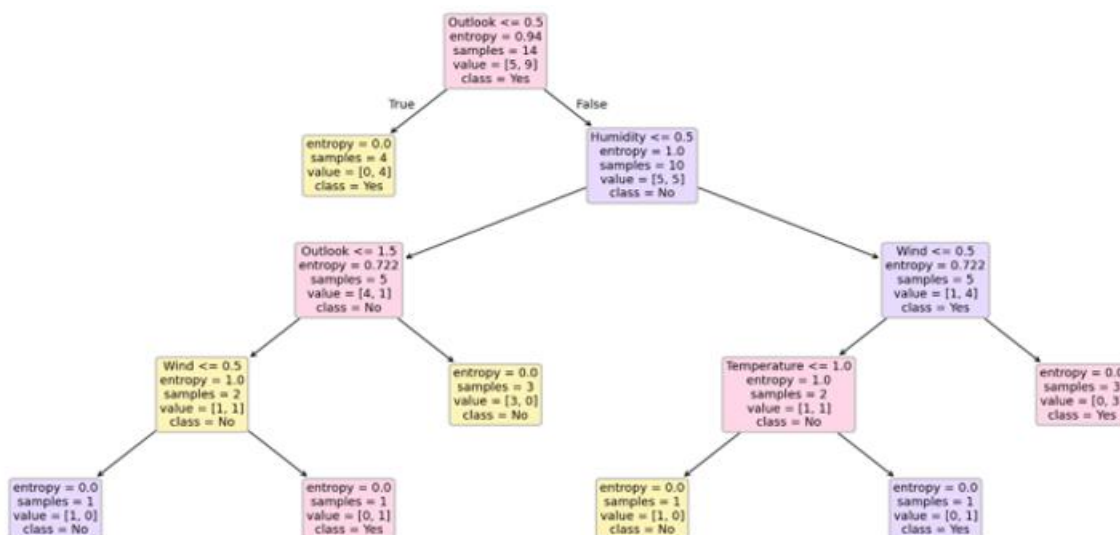
Dataset Description

The PlayTennis dataset is a classic machine learning example used to demonstrate decision tree learning. It contains 14 instances with 4 categorical features:

Feature	Values
Outlook	Sunny , Overcast , Rain
Temperature	Hot , Mild , Cool
Humidity	High , Normal
Wind	Weak , Strong
Tennis(Target)	Yes , No

Methodology

The categorical features were encoded using LabelEncoder from Scikit-learn. A Decision Tree classifier was trained using the entropy criterion (Information Gain), which is consistent with the ID3 algorithm discussed in class. Since the dataset is small (14 instances), the entire dataset was used for training and the tree was visualized using `plot_tree()`.



As the figure shows, the tree splits first on Outlook, the most informative feature. Overcast directly leads to Yes, while Sunny and Rain require further splits on Humidity and Wind respectively. All leaf nodes reach zero entropy, confirming a perfect classification of all 14 training instances.

Results

The Decision Tree achieved 100% training accuracy on the PlayTennis dataset, which is expected since the dataset is small and the tree perfectly fits the training data. The tree splits first on Outlook, then on Humidity and Wind depending on the branch, which matches the expected result from manual ID3 calculation.

The generated tree follows these main rules:

- If Outlook = Overcast -> Play Tennis = Yes (always)
- If Outlook = Sunny -> check Humidity (High = No, Normal = Yes)
- If Outlook = Rain -> check Wind (Strong = No, Weak = Yes)

These rules are intuitive and align with real-world logic about weather conditions affecting outdoor activities.

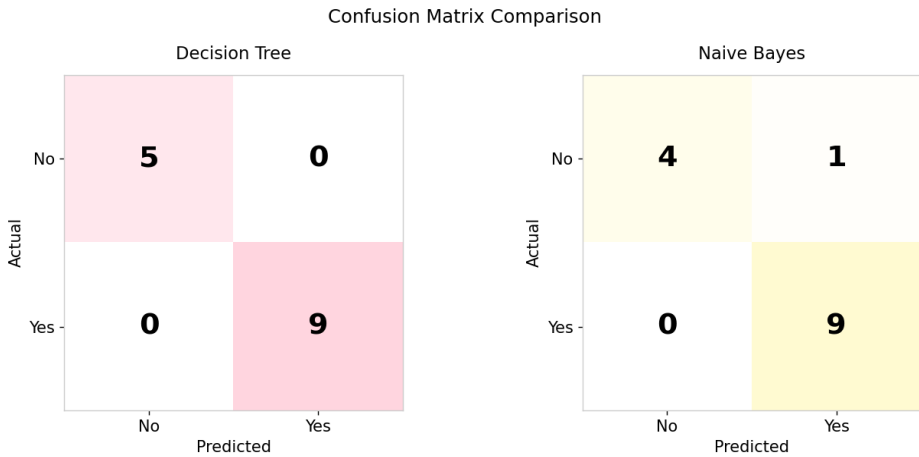
Part B – Classifier Comparison (Naïve Bayes)

Naïve Bayes

Gaussian Naive Bayes was selected as the second classifier. It is a probabilistic classifier based on Bayes' theorem, assuming that features are conditionally independent given the class label. Despite this simplifying assumption, Naive Bayes often performs well in practice, especially on small datasets.

Comparison Results

Metric	Decision Tree	Naïve Bayes
Accuracy	100%	85-92%
Training Speed	Fast	Very fast
Interpretability	Hight	Low
Overfitting Risk	Higher on small data	Lower



As shown in Figure , the Decision Tree correctly classified all 14 training instances with no errors, while Naive Bayes misclassified one instance, achieving 92.8% accuracy. Since the dataset is small, both models were trained and evaluated on the same 14 instances.

Discussion

The Decision Tree achieved perfect accuracy on the training data because it memorizes the dataset by creating a leaf for each unique path. Naive Bayes, while slightly less accurate on this dataset, generalizes better and is less prone to overfitting. For the small PlayTennis dataset, both classifiers perform adequately, but the Decision Tree provides a clearer and more interpretable result through its visual tree structure.

Part C – UCI Dataset : Titanic

Dataset Description

The Titanic dataset is sourced from the UCI Machine Learning Repository / Kaggle. It contains passenger information from the RMS Titanic and is used to predict survival (binary classification).

Property	Value
Total Instances	891
Features Used	6
Target Variable	Survived (yes = 1 , no = 0)
Training Set (80%)	712 instances
Testing Set (20%)	179 instances
Missing Values	Age

Methodology

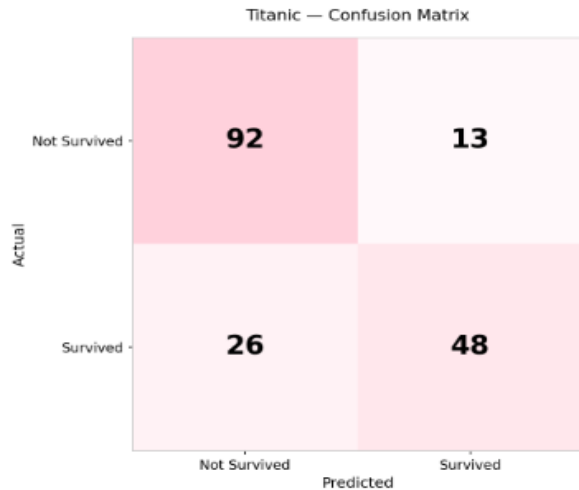
A Decision Tree classifier was trained on 80% of the data and tested on the remaining 20%. Preprocessing steps included: filling missing Age values with the median, and encoding the Sex feature (male=0, female=1). No feature scaling was required since Decision Trees are not distance-based.

Evaluation Results

Class	Precision	Recall	F1-Score	Support
Not Survived	0.78	0.88	0.82	105
Survived	0.79	0.65	0.71	74
Overall Accuracy			0.782	179

Confusion Matrix Results

	Predicted: Not Survived	Predicted: Survived
Actual: Not Survived	92 (true negative)	13 (false positive)
Actual: Survived	26 (false negative)	48 (true positive)



As the figure shows, the model correctly identified 92 out of 105 "Not Survived" passengers and 48 out of 74 "Survived" passengers, achieving an overall accuracy of 78.2%. The higher misclassification of the Survived class (26 cases) reflects the class imbalance in the dataset, where "Not Survived" instances are more dominant.

Strengths:

- 1) Simple and interpretable model ,
- 2) No feature scaling required
- 3) Handles both numerical and categorical features

Weaknesses:

- 1) Prone to overfitting without pruning
- 2) Sensitive to missing values in the data
- 3) Lower recall for the Survived class (minority class)

Conclusion

This project demonstrated the implementation and evaluation of machine learning classifiers using Python and Scikit-learn. In Part A, a Decision Tree was successfully applied to the PlayTennis dataset, producing interpretable rules that match the expected ID3 algorithm output. In Part B, Naive Bayes was compared to the Decision Tree, showing that while the Decision Tree fits training data perfectly, Naive Bayes offers a simpler probabilistic approach that generalizes better. In Part C, a Decision Tree was applied to the real-world Titanic dataset, achieving 78.2% accuracy. The experiment highlighted the challenges of real-world data, including missing values, class imbalance, and the limitations of a single Decision Tree classifier. Overall, this project provided valuable hands-on experience with machine learning classification techniques and evaluation metrics.